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Causal restriction is not spectral truncation: a no-go result for the Rindler–Casimir mechanism of quantised inertia

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ABSTRACT

Quantised inertia proposes that acceleration-dependent causal horizons modify inertia, in one formulation through an asymmetry in the momentum flux attributed to Unruh radiation. We test that mechanism at the quantum-field-theoretic step on which it depends. In algebraic quantum field theory an acceleration horizon imposes no reflecting or Casimir boundary condition: the Minkowski vacuum is restricted to the Rindler-wedge algebra, where it is a KMS state under boost modular flow. Restricting a state to a subalgebra leaves unchanged the expectation value of every local observable supported there, so the renormalised vacuum stress tensor is unaltered. Poincaré invariance fixes that tensor to be isotropic, and conventionally zero, and isotropy is a tensor statement that survives passage to an accelerated frame. The one-sided mean radiation-pressure deficit the mechanism requires is therefore not generated by the horizon. The natural replies, including the Fulling–Rindler vacuum, detector back-reaction and a horizon-limited uncertainty relation, do not restore it without further physical input. We further note that the cosmological scale Θ denotes inequivalent geometrical objects in this literature, implying different critical-acceleration histories not, to our knowledge, tested in common. An appendix corrects the discrete mode count in Renda's published sceptical analysis: restoring the polarisation multiplicities of a conducting cavity removes the reported $F \simeq 2.17$ overshoot and the stable acceleration inferred from it, and a factor-of-two convention in the largest admissible wavelength reproduces the abscissa of the published critical point. Neither correction weakens that paper's two principal criticisms.

Key words: gravitation – methods: analytical – galaxies: kinematics and dynamics – cosmology: theory – dark matter

1 INTRODUCTION

The mathematical statement of inertia is elementary; its physical origin is less so. This has repeatedly motivated attempts to connect inertial reaction to vacuum structure, Machian information, acceleration horizons or cosmological boundary conditions. Among the most explicit is Quantised Inertia, developed by McCulloch McCulloch (2007, 2013, 2016), in which acceleration-dependent horizons are assigned a role in modifying the inertial response of matter.

The proposal has an unusual virtue: it attempts to name a microscopic mechanism rather than merely replacing Newtonian dynamics with a phenomenological interpolation law. Uniform acceleration certainly does introduce a distinguished causal horizon, and quantum field theory certainly associates non-trivial thermodynamic structure with the corresponding wedge. The Unruh effect (Unruh 1976) and the Bisognano–Wichmann theorem (Bisognano & Wichmann 1976) make that statement precise.

It is therefore possible to ask a sharper question than whether a particular Quantised-Inertia formula fits a particular set of observations:

Does the quantum-field-theoretic structure associated with an acceleration horizon actually perform the operation required by the proposed inertial mechanism?

For the asymmetric Rindler–Casimir mechanism, the answer is no. The central distinction is

restriction of the observable algebra
$\neq$
restriction of the allowed field configurations or modes.

(1)

A material boundary condition, compactification, topological identification or genuine change of quantum state may alter the field problem and its stress tensor. A Rindler horizon in the Minkowski vacuum does not do so merely by appearing in the causal description of an accelerated observer.

Previous criticism has already identified difficulties internal to the Quantised-Inertia calculation. Renda (2019), in particular, reanalysed the discrete black-body sampling used in the 2007 and 2013 formulations and found substantial problems in the spectral sampling and in the use of the peak Unruh wavelength as a proxy for the energy density. In reproducing that calculation we also find a discrete mode-counting prob-

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lem in the treatment of zero-index cavity modes: the use of OEIS A002102 together with a universal factor of two for polarisation overcounts precisely the boundary modes that dominate the small-p regime. Restoring the Maxwell polarisation multiplicities removes the reported $F \simeq 2.17$ overshoot and the associated stable acceleration; Appendix A gives the calculation. This correction does not affect Renda's two principal criticisms. The present argument attacks an earlier logical step: whether a causal horizon supplies the cavity-like operation that any such spectral calculation presupposes in the first place.

We emphasise equally what is not claimed. This paper does not show that every empirical formula proposed under the name Quantised Inertia must fail observationally. It does not disprove MOND or other modified-inertia phenomenology. Nor does it establish that inertia cannot have a vacuum, causal, informational or modular origin by another route. It shows that one proposed route does not follow from standard quantum field theory.

2 THE PROPOSALS, DISTINGUISHED

Quantised Inertia has appeared in several formulations. They share a family resemblance but are not logically equivalent and should not be treated as a single target.

2.1 Cosmological mode restriction

In the earlier formulation (McCulloch 2007), Unruh wavelengths too large to be adequately represented within a cosmological scale are assumed to be poorly sampled or excluded. Schematically,

$$a \longrightarrow \lambda_{\mathrm{U}}(a) \longrightarrow \text{cosmological spectral restriction} \longrightarrow m_{\mathrm{I}}(a). \tag{2}$$

The operative physical assumption is therefore stronger than the Unruh effect itself: a causal or cosmological length is assumed to constrain the field spectrum in a cavity-like manner.

The characteristic acceleration obtainable from c and a cosmological length L is

$$a_H \sim \frac{c^2}{L}. \tag{3}$$

Dimensional agreement with other low-acceleration scales is interesting but does not by itself establish the microscopic mechanism.

2.2 Asymmetric Rindler suppression

The later formulation (McCulloch 2013) makes the proposed reaction local. An accelerating body possesses a Rindler horizon behind it; the horizon is taken to reduce the Unruh radiation available from that direction relative to the opposite direction, producing an asymmetric pressure that opposes acceleration:

$$\text{acceleration} \longrightarrow \text{Rindler horizon} \longrightarrow \Delta P_{\mathrm{U}} \longrightarrow F_{\mathrm{I}}. \tag{4}$$

McCulloch expresses the force in terms of directional Unruh energy densities and the radiation-pressure relation $P = u/3$. In local QFT the corresponding mean normal pressure on a surface with spacelike unit normal n^μ is encoded in

$$P_n \doteq \langle T_{\mu\nu} \rangle \cdot n^\mu n^\nu. \tag{5}$$

Thus a one-sided mean pressure deficit of the proposed kind requires a directional asymmetry in the expectation value of the local stress energy.

2.3 Uncertainty-principle reformulation

A subsequent reformulation (McCulloch 2016) begins from additional assumptions involving an uncertainty energy and a reduction of position uncertainty by relativistic horizons. A characteristic horizon distance

$$\Delta x \sim \frac{c^2}{a} \tag{6}$$

is then used to generate an acceleration-dependent energy scale. This construction is logically distinct from (4). Reproducing a similar acceleration dependence does not establish the radiation-pressure mechanism, and the uncertainty formulation is therefore considered separately in Section 10.

3 WHAT A RINDLER HORIZON ACTUALLY DOES

Let ω_0 denote the Minkowski vacuum, let W be the right Rindler wedge, and let $\mathcal{A}(W)$ denote the algebra of observables localised in W (Haag 1996). The state available on that algebra is the restriction

$$\omega_W \doteq \omega_0\big|_{\mathcal{A}(W)}. \tag{7}$$

Nothing in (7) changes the global field state. The same state functional is simply being evaluated on a smaller algebra.

The Bisognano–Wichmann theorem (Bisognano & Wichmann 1976) identifies the modular automorphism group of the wedge vacuum with the Lorentz boosts preserving W. Transcribed into the proper time of a uniformly accelerated detector with proper acceleration a, the corresponding KMS temperature is

$$T_{\mathrm{U}} = \frac{\hbar a}{2\pi c \cdot k_B}. \tag{8}$$

Thus the thermal structure belongs to the pair

$$\big(\mathcal{A}(W), \omega_0|_{\mathcal{A}(W)}\big). \tag{9}$$

It is not produced by imposing a reflecting condition on the Rindler horizon, replacing the Minkowski vacuum by a different vacuum state, or removing global solutions of the field equation. The field remains an ordinary Minkowski field; the observer has lost access to part of its algebra.

4 THE NO-GO RESULT

Proposition 1 (Restriction changes access, not local expectation values)**.** *Let ω be a state on an algebra $\mathcal{A}$ and let $\mathcal{B} \subset \mathcal{A}$ be a subalgebra. Then*

$$\omega|_{\mathcal{B}}(A) = \omega(A) \qquad \forall A \in \mathcal{B}. \tag{10}$$

Proof. This is the definition of restriction of a functional to a subalgebra. □

Proposition 1 is mathematically elementary. Its physical consequence here is not. Restricting the Minkowski vacuum to the wedge changes the modular structure of the state and the set of jointly accessible observables; it does not manufacture new local expectation values inside the wedge.

4.1 The local stress tensor

The mechanism (4) requires an asymmetric local mean momentum flux. The appropriate local quantity is therefore the renormalised stress-energy tensor.

Because $T_{\mu\nu}(x)$ is an operator-valued distribution, it is cleaner to work with a smeared local field. For $f \in C_0^\infty(W)$, write

$$T_{\mu\nu}(f) \doteq \int d^4x\, f(x) \cdot T_{\mu\nu}(x). \tag{11}$$

In the local-covariant formulation the renormalised stress tensor is a local covariant quantum field (Hollands & Wald 2001). Its expectation value for $\operatorname{supp} f \subset W$ is therefore determined by the state restricted to the wedge. Hence

$$\langle T_{\mu\nu}(f)\rangle_{\omega_W} = \langle T_{\mu\nu}(f)\rangle_{\omega_0}. \tag{12}$$

Poincaré invariance restricts the renormalised Minkowski-vacuum expectation value to a multiple of the metric,

$$\langle T_{\mu\nu}(x)\rangle_{\omega_0} = C \cdot \eta_{\mu\nu}, \tag{13}$$

and the conventional Minkowski normalisation sets $C = 0$. The argument below does not need that choice. For any value of C, the mean normal pressure (5) on a surface with spacelike unit normal n^μ is

$$P_n = C \cdot n^\mu n_\mu, \tag{14}$$

which has the same value for every spacelike unit normal n^μ, whatever the metric signature convention. The forward and rearward normal pressures are therefore equal and their difference vanishes.

There is thus no *anisotropic* Minkowski-vacuum mean stress created merely by restricting causal access to W, and this conclusion is independent of the additive vacuum normalisation. With the conventional choice $C = 0$ the mean stress vanishes outright, but the no-go result below rests only on isotropy.

4.2 The absence of directional stress is frame-independent

Equation (13) is a tensor statement, not a statement tied to an inertial coordinate system. Under any regular coordinate transformation

$$\langle T'_{\alpha\beta}\rangle = \frac{\partial x^\mu}{\partial x'^\alpha}\frac{\partial x^\nu}{\partial x'^\beta}\langle T_{\mu\nu}\rangle = C \cdot g'_{\alpha\beta}. \tag{15}$$

Equivalently, contracting with any orthonormal tetrad e^μ_a carried by the accelerated observer gives $\langle T_{ab}\rangle = C \cdot \eta_{ab}$: the off-diagonal components, and hence the mean local momentum flux, vanish, and the three principal pressures are equal. With the conventional choice $C = 0$ every component vanishes separately; for any other choice the mean stress measured in the accelerated frame remains isotropic.

This closes the natural retreat that the mean momentum flux required by (4) exists only in the accelerated frame. Detector response is observer dependent; the isotropy of the Minkowski-vacuum mean stress is not.

4.3 No-go theorem for the asymmetric pressure mechanism

Corollary 1 (Rindler–Casimir no-go)**.** *Consider standard quantum field theory on Minkowski spacetime, with the field in the Minkowski vacuum. Suppose an inertial reaction is attributed solely to an asymmetry in the expectation value of the local field stress produced by an accelerated body's Rindler horizon. Then that proposed reaction vanishes: causal restriction to the wedge does not generate the required asymmetric mean momentum flux. The conclusion is independent of the additive normalisation of the vacuum stress tensor: no anisotropic pressure arises for any value of C in (13).*

The assumptions in Corollary 1 are deliberate. If a model adds new interactions, changes the quantum state, introduces a material boundary, modifies the field equations, or postulates an additional non-QFT response to causal horizons, then it has added physics beyond the mechanism addressed here. Such additions may be studied independently, but they cannot be presented as consequences of the ordinary Unruh/Bisognano–Wichmann structure alone.

The operation provided by standard QFT is

$$\boxed{\begin{aligned}\text{Rindler horizon} &\longrightarrow \text{restriction of accessible algebra}\\ &\longrightarrow \text{modular/KMS structure,}\end{aligned}} \tag{16}$$

not

$$\boxed{\begin{aligned}\text{Rindler horizon} &\longrightarrow \text{one-sided removal of field modes}\\ &\longrightarrow \Delta P.\end{aligned}} \tag{17}$$

5 WHY THIS IS NOT A CASIMIR BOUNDARY

A Casimir system and a Rindler wedge represent different physical operations. A material boundary, imposed boundary condition, compactification or non-trivial global identification modifies the field problem. The allowed solutions or the global state may consequently differ from those of unconstrained Minkowski space, and a non-zero change

$$\Delta\langle T_{\mu\nu}\rangle \tag{18}$$

may result.

An observer-dependent causal horizon in Minkowski spacetime imposes no Dirichlet, Neumann, Robin, conducting, periodic or analogous boundary condition on the field. No plate appears at the horizon and the field is regularly defined across it in the global Minkowski description.

Calling the Rindler horizon itself an asymmetric Casimir boundary therefore attributes to causal inaccessibility an operation that the field theory does not perform.

6 THE FULLING–RINDLER-VACUUM RETREAT

A natural objection is that the relevant state should be the Fulling–Rindler vacuum ω_{R} rather than the Minkowski vacuum restricted to the wedge.

This objection must be treated carefully because the

Fulling–Rindler vacuum does *not* require a mirror at the horizon. Rindler spacetime may be treated as a self-contained globally hyperbolic static spacetime, and the Fulling–Rindler quantisation is defined without imposing a reflecting boundary condition at the horizon (Fulling 1973; Fulling & Unruh 2004). The relevant distinction is one of states.

6.1 It is a different quantum state

The Minkowski vacuum ω_0 and the Fulling–Rindler vacuum ω_{R} are distinct states. The ordinary Unruh problem considers a detector or body accelerated through the Minkowski vacuum. Changing the trajectory of that body does not, by itself, replace the field state by ω_{R}.

Consequently,

$$\langle T_{\mu\nu}\rangle_{\omega_{\mathrm{R}}} - \langle T_{\mu\nu}\rangle_{\omega_0} \neq 0 \tag{19}$$

demonstrates state dependence; it does not demonstrate that restriction of ω_0 to a wedge generated a new local stress.

If the asymmetric-pressure mechanism requires the Fulling–Rindler stress rather than the restricted Minkowski state, an additional dynamical step is required:

$$\boxed{\text{acceleration through } \omega_0 \;\longrightarrow\; \text{preparation of } \omega_{\mathrm{R}}} \tag{20}$$

or of some other non-Minkowski state. The existence of the causal horizon supplies no such state-preparation law.

6.2 It is not regular across the Minkowski horizon

The Fulling–Rindler vacuum has singular behaviour as the Rindler horizon is approached and does not represent a regular vacuum state on all of Minkowski spacetime across that horizon (Candelas & Deutsch 1977; Fulling & Unruh 2004). This is not an objection to using it as a state intrinsic to Rindler spacetime. It is precisely why it must not be confused with the Minkowski vacuum seen by an accelerated observer in globally empty flat spacetime.

6.3 What a real accelerated boundary would change

A physical accelerated mirror or cavity is another problem again. It genuinely modifies the field by imposing boundary conditions and can generate boundary-dependent stresses. In that case the physical boundary has been supplied by matter. The effect is not generated by causal inaccessibility alone.

Remark 1. The three constructions

$$\begin{cases} \text{restricted Minkowski state,} \\ \text{Fulling–Rindler state,} \\ \text{mirror-bounded accelerated system} \end{cases}$$

are mutually distinct.

7 UNRUH THERMALITY IS NOT A PRE-EXISTING RADIATION BATH

An Unruh–DeWitt detector following a uniformly accelerated worldline through the Minkowski vacuum has a thermal response at (8). This is encoded in the pullback of the vacuum correlation function to the accelerated trajectory and, algebraically, in the KMS structure of the wedge vacuum under boosts (Unruh 1976; Crispino et al. 2008).

It does not follow that the Minkowski vacuum contains a classical local thermal fluid with stress tensor $T_{\mu\nu}^{\mathrm{bath}}$ whose radiation can simply be blocked on one side. The simultaneous statements

$$\begin{cases} T_{\mathrm{detector}} = T_{\mathrm{U}}, \\ \langle T_{\mu\nu}\rangle_{\omega_0} = C \cdot \eta_{\mu\nu} \end{cases} \tag{21}$$

are not contradictory. They concern different observables.

A derivation of a mechanical pressure from Unruh thermality must therefore provide a dynamical bridge from detector response or field fluctuations to a universal directed force. Causal restriction alone is not that bridge.

8 GENUINE DETECTOR RESPONSE AND BACK-REACTION

It would be incorrect to infer from the absence of an anisotropic Minkowski-vacuum mean stress that an accelerated system coupled to a quantum field can never exchange energy or momentum with it. It can.

Unruh & Wald (1984) analysed a uniformly accelerated two-level detector coupled to a quantum field and showed explicitly how a process described as absorption of a Rindler particle in the accelerated description is represented in the inertial description as emission of a Minkowski quantum. The apparent puzzles of causality and energy bookkeeping are resolved within the interacting detector–field system.

Raine et al. (1991), studying a uniformly accelerated quantum oscillator, further showed that a naive identification of Unruh response with ordinary emitted radiation is unreliable once the relevant quantum correlations are retained.

These results do not reconstruct the mean-pressure mechanism of (4). Three distinctions matter.

8.0.0.1 The response depends on interaction. Detector excitation, dissipation and radiation depend on a specified interaction Hamiltonian. In the zero-coupling limit that detector response disappears.

8.0.0.2 Detector response is not a pre-existing vacuum pressure. An interacting detector may perturb the field and exchange energy or momentum with it. This does not imply that the uncoupled Minkowski vacuum possessed the asymmetric local mean stress required by (4).

8.0.0.3 A universal inertial law has not been derived. Nothing in the detector analysis by itself yields, for arbitrary matter, a universal body-independent reaction

$$F_{\mathrm{I}} = -m \cdot a. \tag{22}$$

Obtaining such a law would require an additional derivation relating specific detector–field dynamics to inertial mass.

Genuine accelerated-detector response therefore exists, but its existence does not repair the missing step in the asymmetric-Casimir pressure mechanism.

9 THE COSMOLOGICAL VERSION

The Rindler no-go theorem above is a statement about the Minkowski vacuum and should not be silently promoted to a theorem about arbitrary cosmological states. The earlier Hubble-scale construction is instead constrained by a more elementary distinction.

A cosmological or causal horizon may alter causal accessibility and may carry thermodynamic significance. It does not follow merely from the existence of that horizon that wavelengths longer than some chosen scale are excluded as though the field had been placed in a material cavity.

A spectral selection rule requires an independent derivation from

$$\begin{cases} \text{field boundary conditions,} \\ \text{global topology,} \\ \text{global quantum state,} \\ \text{modified dynamics.} \end{cases} \tag{23}$$

Thus

$$\boxed{\text{horizon length} \not\Rightarrow \text{Casimir spectrum}} \tag{24}$$

without further physical input.

9.1 The moving target Θ

There is a second cosmological issue which is logically independent of the spectral boundary objection. The quantity denoted by Θ is not one geometrically fixed object throughout the canonical Quantised-Inertia literature. The original formulation identifies a limiting scale with twice the Hubble distance (McCulloch 2007); if that language is read literally at a general epoch, one obtains

$$\Theta_{\mathrm{H}}(t) \doteq \frac{2c}{H(t)}. \tag{25}$$

The same paper already recognises that cosmic evolution of the limiting scale would make the associated minimum acceleration evolve and notes that the effect could become much larger in the early Universe.

The later galactic formulation (McCulloch 2017) instead adopts the present value

$$\Theta_0 \doteq 8.8 \times 10^{26}\,\mathrm{m} \tag{26}$$

and calls it the "co-moving cosmic diameter", taken from an estimate of the present diameter of the observable universe and explicitly distinguished from the 2.6×10^{26} m value of the earlier work (McCulloch 2007, 2012), which is there glossed as the cosmic diameter at the epoch of emission (McCulloch 2017). The same paper writes Θ as the Hubble diameter where the inertial mass is defined, while its abstract and its numerical section identify Θ with the co-moving diameter of the observable universe. Those are not the same quantity, and $2c/H_0$ and 8.8×10^{26} m differ by a factor of roughly 3.4. For earlier epochs the same paper then prescribes, on the stated grounds of assuming a linear expansion of the cosmos,

$$\Theta_{\mathrm{QI}}(z) \doteq \frac{\Theta_0}{1+z}. \tag{27}$$

The stated justification does not in fact single out (27): the proper length of a fixed comoving interval scales as $(1+z)^{-1}$ whatever the expansion history, so no linearity assumption is needed to obtain it; conversely, a genuinely co-moving diameter does not evolve at all, and it is the corresponding proper scale that varies. Which of the two is intended matters, because only one of them enters a local dynamical law. Later applications again use the fixed present-epoch 8.8×10^{26} m scale, without redshift correction (McCulloch & Lucio 2019; McCulloch 2024). The terminology is therefore somewhat treacherous. A Hubble diameter, a particle-horizon diameter, an event-horizon diameter, and the physical size at redshift z of a fixed present comoving interval are inequivalent functions of cosmic time. Schematically,

$$\Theta(t) \in \begin{cases} \dfrac{2c}{H(t)} & \text{Hubble diameter,} \\ 2a_{\mathrm{c}}(t) \cdot \displaystyle\int_0^t \frac{c \cdot dt'}{a_{\mathrm{c}}(t')} & \text{particle-horizon diameter,} \\ 2a_{\mathrm{c}}(t) \cdot \displaystyle\int_t^\infty \frac{c \cdot dt'}{a_{\mathrm{c}}(t')} & \text{event-horizon diameter,} \\ \dfrac{\Theta_0}{1+z} & \text{2017 QI prescription,} \\ \Theta_0 & \text{fixed present-epoch proxy.} \end{cases} \tag{28}$$

Here $a_{\mathrm{c}}(t)$ denotes the cosmological scale factor, to avoid confusion with a local acceleration; the event-horizon entry presumes that the corresponding integral is finite. These alternatives agree neither numerically nor dynamically in a general cosmology.

9.1.0.1 The empirical question remains open. The ambiguity in Θ should not be read merely as another criticism of Quantised Inertia. It also defines an empirical discrimination programme. McCulloch has repeatedly confronted particular formulations of the theory with observation, including present-epoch galaxy rotation, high-redshift galaxies, wide binaries and, more recently, the Alpha–Proxima Centauri system (McCulloch 2017; McCulloch & Lucio 2019; McCulloch 2024). The high-redshift comparison is particularly relevant here because it already treats an evolving cosmic scale as an observational prediction rather than as a bookkeeping correction. What appears to be absent from the published programme is a common test in which the competing geometrically defensible definitions of $\Theta(t)$ are each evolved consistently through the same cosmology and confronted, without redefinition or epoch-by-epoch retuning, with the same independent body of observations.

That absence should not be read as a failure on McCulloch's part. An exhaustive multi-observable cosmological test is a substantial programme and cannot reasonably be expected of one proponent, particularly for a speculative theory that has attracted limited independent development. It nevertheless leaves an experimentally accessible question unresolved. The no-go result of the present paper concerns the proposed Rindler–Casimir mechanism; it does not logically imply that every numerical relation associated with Quantised Inertia must fail in nature. Were one consistently defined and evolved choice of Θ unexpectedly to reproduce galaxy dynamics across redshift and then survive independent cosmological

tests without further adjustment, that would constitute important empirical evidence even though the mechanism advanced to explain it here has failed. The scientific task would then be to reconstruct a viable physical mechanism capable of producing the successful phenomenology. Conversely, if no defensible definition of $\Theta(t)$ survived a common test, the empirical case for the phenomenology itself would be correspondingly weakened. Either outcome would be more informative than leaving the alternatives untested.

That ambiguity is consequential because the later Quantised-Inertia formulae contain the characteristic acceleration

$$a_\Theta(t) \doteq \frac{2c^2}{\Theta(t)}. \tag{29}$$

For two of the prescriptions above,

$$a_\Theta = \begin{cases} c \cdot H(t), & \Theta(t) \doteq 2c/H(t), \\ a_{\Theta,0} \cdot (1+z), & \Theta(z) \doteq \Theta_0/(1+z). \end{cases} \tag{30}$$

The low acceleration correction is consequently not a universal constant carried unchanged through cosmic history. Its magnitude depends on which geometrical quantity Θ is meant to represent and on the evolution law assigned to that quantity.

9.2 Time-dependent and interaction-dependent inertia

The dynamical consequences become more severe when the interaction-dependent reading of Quantised Inertia is taken literally. The reference acceleration has itself become progressively more specific in the later literature. In the galactic formulation it is described as acceleration relative to surrounding matter (McCulloch 2017); the wide-binary analysis invokes acceleration relative to the fixed stars while explicitly rejecting a MOND-like external-field effect (McCulloch & Lucio 2019); and the Alpha–Proxima analysis states directly that the inertial mass for each interaction is determined only by the mutual acceleration in that interaction (McCulloch 2024). Taking this last prescription literally, for a pair i, j at cosmic time t,

$$m_{\mathrm{I}}^{(ij)}(t) \doteq m_{\mathrm{g}}^{(i)} \left(1 - \frac{2c^2}{|a_{ij}(t)| \cdot \Theta(t)}\right). \tag{31}$$

A many-body system then carries not one modified inertial mass for body i, but an interaction-indexed family

$$\left\{ m_{\mathrm{I}}^{(ij)}(t) \right\}_{j \neq i}. \tag{32}$$

Even before asking how non-gravitational interactions are to be treated, a cosmological many-body theory must therefore specify how simultaneous forces are to compose when the inertial response associated with each interaction is different and all of those responses evolve through $\Theta(t)$. The equations of motion therefore become coupled and implicit, with additional explicit cosmological time dependence if $\Theta = \Theta(t)$, unless a more fundamental dynamical prescription is supplied.

This is also where the conservation question enters. It would be misleading simply to say that a time-dependent Θ "violates energy conservation". Noether's theorem states a conditional result: a conserved energy follows from time-translation invariance of an action (Noether 1918). If a putative mechanical Lagrangian depends on an externally prescribed $\Theta(t)$, then that symmetry is absent. If the explicit time dependence enters only through Θ, one has

$$\begin{cases} E \doteq \displaystyle\sum_A \dot{q}^A \cdot \frac{\partial L}{\partial \dot{q}^A} - L, \\ \dfrac{dE}{dt} = -\dfrac{\partial L}{\partial t} = -\dfrac{\partial L}{\partial \Theta} \cdot \dot{\Theta}. \end{cases} \tag{33}$$

Thus there is no reason for the ordinary mechanical energy to remain constant. If no action is supplied at all, the corresponding Noether bookkeeping cannot yet be posed.

Nor is global energy conservation available in this naive form in an arbitrary cosmological spacetime. General relativity does not in general furnish a globally conserved energy in the absence of the appropriate spacetime symmetry. What it does furnish is a covariant field theory in which the contracted Bianchi identity and Einstein equation imply local covariant stress-energy conservation (Wald 1984):

$$\left.\begin{aligned} \nabla_\mu G^{\mu\nu} &= 0, \\ G^{\mu\nu} &= \frac{8\pi G}{c^4} \cdot T^{\mu\nu} \end{aligned}\right\} \quad \Longrightarrow \quad \nabla_\mu T^{\mu\nu} = 0. \tag{34}$$

The point is therefore not that Quantised Inertia must conserve a Newtonian energy which even general relativity does not generally possess. It is that a cosmological completion must provide its own consistent covariant bookkeeping. One possibility would be to promote the quantity represented by Θ to part of the dynamical system, so that energy-momentum exchange with that sector is derived rather than inserted as explicit time dependence. Other completions are possible; none follows from the present phenomenological prescription alone.

These distinctions potentially have observational consequences far beyond the present-day low-acceleration systems usually considered. McCulloch already identified redshift-dependent galaxy dynamics as a prediction of an evolving Θ (McCulloch 2007, 2017). More generally, an epoch-dependent inertial response could affect

$$\begin{cases} \text{high-redshift rotation curves and velocity dispersions,} \\ \text{virialisation and the growth of cosmic structure,} \\ \text{early-time baryonic dynamics,} \\ \text{potentially the acoustic dynamics imprinted in the CMB.} \end{cases} \tag{35}$$

The last three statements are consistency targets rather than predictions calculated here. Quantised Inertia as presently formulated does not supply the covariant many-body and cosmological perturbation theory required to compute them. The absence of such a calculation prevents these observations from being used here as a further no-go result, but it also prevents the cosmological evolution of Θ from being treated as an innocuous detail.

A phenomenological weighting function suppressing Unruh wavelengths according to their relation to a cosmological scale may of course be postulated. It is then an additional hypothesis, not a consequence of the Unruh or Bisognano–Wichmann results; and if the scale is allowed to evolve, its geometrical meaning and dynamics must be specified as part of that hypothesis.

10 THE UNCERTAINTY-PRINCIPLE FORMULATION

The uncertainty-principle formulation (McCulloch 2016) does not require a literal local radiation bath and therefore is not directly excluded by Corollary 1. It does, however, require independent assumptions.

First, the geometrical distance to a causal horizon is identified with a position uncertainty. A classical horizon distance

$$d_H \sim \frac{c^2}{a} \tag{36}$$

is not automatically the variance

$$\Delta x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2} \tag{37}$$

of a specified quantum state. The state and observable for which such an identification holds must be supplied.

Second, an uncertainty relation constrains dispersions:

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}. \tag{38}$$

It does not by itself constitute an equation of motion or a conservation law converting a change of uncertainty into a mechanical force. McCulloch's 2016 construction explicitly introduces additional conservation and horizon–uncertainty assumptions. Those assumptions may be investigated on their own terms, but they are not consequences of the Rindler–Casimir mechanism.

Accordingly, the uncertainty formulation may reproduce a characteristic acceleration dependence while remaining logically independent of the mechanism ruled out above.

11 RELATION TO PREVIOUS CRITICISM

Renda (2019) subjected the 2007 and 2013 derivations of Quantised Inertia to a sceptical analysis. His objections concern the calculation after a cavity-like spectral restriction has already been assumed: the discretisation of the black-body spectrum and the use of the peak wavelength as a proxy for the total energy density.

There is, however, a further correction to the discrete calculation. Renda first labels cavity modes by non-negative integers and then, in his eq. (14), supplies a factor of two for the two electromagnetic polarisations. His eq. (16) subsequently takes the continuum density to be the volume of one octant, explicitly writing $l, m, n > 0$, while the discrete sum uses OEIS A002102, which counts all non-negative triples. The difference is negligible in the continuum limit but not at the small mode numbers responsible for the reported low-acceleration peak. More importantly, the boundary lattice points do not carry the same polarisation multiplicity as interior points. For a rectangular perfectly conducting cavity, a mode with all three indices non-zero has two independent polarisations; a mode with exactly one zero index has one; and a putative mode with two zero indices has vanishing field amplitude. If $N_3(s)$ counts strictly positive ordered triples whose squares sum to s and $N_2(s)$ counts strictly positive ordered pairs, the physical electromagnetic degeneracy is therefore

$$g(s) \doteq 2N_3(s) + 3N_2(s), \tag{39}$$

not twice A002102(s).

Re-evaluating Renda's eq. (28) with (39) removes the $F \simeq 2.17$ overshoot and the inferred stable acceleration: over the low-acceleration window containing the reported critical point the correctly weighted result rises towards the continuum limit without an interior maximum. Appendix A gives the counting and a numerical reproduction of both results. This correction does not weaken Renda's central conclusions: McCulloch's linear interpolation is still not a derivation, and the peak-wavelength substitution remains independent of the mode-counting issue.

The present result addresses the preceding question:

what physical operation makes a Rindler horizon a cavity?	(40)

Under standard Minkowski QFT, causal restriction does not do so. Renda's criticism and the present one are therefore complementary rather than redundant.

12 WHAT THE RESULT DOES NOT SHOW

The scope of the result should be stated exactly.

(i) Nothing here proves that every empirical formula proposed under the name Quantised Inertia must fail observationally.

(ii) Nothing here proves that inertia cannot ultimately have a vacuum, causal, informational or modular origin.

(iii) Nothing here constitutes a criticism of MOND as such. MOND may be analysed independently of the microscopic mechanism examined here.

(iv) The uncertainty-principle construction of Section 10 is not logically falsified by the Rindler–Casimir no-go theorem. It is shown instead to require independent assumptions.

(v) A theory containing genuinely new horizon-dependent dynamics is not excluded. Such a theory would, however, no longer derive the required pressure from the standard Unruh/Bisognano–Wichmann structure alone.

What has been established is narrower:

Within standard quantum field theory in Minkowski spacetime, causal restriction by an acceleration horizon does not generate the asymmetric local mean vacuum stress required by the proposed Rindler–Casimir inertial mechanism.	(41)

13 WHAT SURVIVES

There is nevertheless a mathematically meaningful residue of the original intuition. Acceleration really does alter the causal decomposition relevant to an observer, and causal restriction really does uncover non-trivial quantum structure. The mistake is to represent that structure as deletion of ordinary field modes.

The natural language is instead modular theory. A global state restricted to a local algebra determines modular flow and KMS structure; for suitable families of regions, additional geometric constructions such as modular connections and their holonomies may be defined (Czech et al. 2018, 2023).

The gauge-invariant information is carried by the appropriate conjugacy class of the holonomy rather than by a choice of modular frame.

This suggests a different question: whether global spacetime structure can modify modular transport of finite causal domains even when their local geometry and local observable algebra agree with those of a universal cover. That question requires no identification of a horizon with a Casimir boundary and no assumption of an asymmetric vacuum pressure. It is logically independent of the mechanism ruled out here.

14 CONCLUSION

Quantised Inertia starts from a legitimate physical observation: acceleration changes the causal organisation relevant to an observer, and quantum field theory assigns non-trivial thermodynamic structure to that change.

The failure occurs in the translation from causal structure to field dynamics. For a body accelerated through the Minkowski vacuum, the Rindler horizon does not impose a reflecting boundary condition, does not replace the Minkowski vacuum by the Fulling–Rindler vacuum, and does not generate an anisotropic local mean vacuum stress by restriction alone. The Minkowski vacuum restricted to the wedge is KMS with respect to boost modular flow while its local renormalised mean stress remains that of the Minkowski vacuum.

The central distinction is therefore

$$\boxed{\text{causal restriction is not spectral truncation.}} \tag{42}$$

The Fulling–Rindler vacuum does not evade this conclusion because it is a different state, not the state dynamically produced merely by accelerating a body through empty Minkowski spacetime. Accelerated detector–field systems do exhibit genuine quantum response and back-reaction, but those effects depend on specified interactions and do not themselves furnish the one-sided mean pressure assumed in the 2013 mechanism or a universal inertial law for arbitrary matter. The uncertainty-principle formulation likewise introduces assumptions independent of that pressure mechanism.

Thus Quantised Inertia, *as hypothesised through the asymmetric Rindler–Casimir mechanism*, is dead on arrival: the causal horizon does not perform the quantum-field-theoretic operation that the proposed force requires.

What remains open is the broader question that motivated it: whether causal or modular structure participates in inertia through some different mechanism.

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DATA AVAILABILITY

No observational data were used in this work. The numerical values in Appendix A are generated directly from the equations, parameters and integer mode counts stated there, and are reproducible in full from that material; the evaluation script is available from the author on request.

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APPENDIX A: ON THE MODE-COUNTING FUNCTION IN RENDA (2019)

The central conclusions of the sceptical analysis by Renda (2019) are not in question here. This appendix concerns the discrete mode count used in its §3.2–3.3. The issue is small in the continuum limit but large at the low mode numbers that produce the reported peak in $F(a)$.

The framing should be stated plainly, because the main text of this paper argues that the cavity in question does not exist. What follows is an internal consistency check, conducted *arguendo*: Renda's calculation is a conducting-cavity calculation, and if that idealisation is granted for the sake of argument then the mode count must be the conducting-cavity one. We are not endorsing the idealisation. The correction below is also robust to that reservation, since the two defensible conventions — strictly interior modes, or interior modes plus correctly weighted boundary modes — remove the reported peak alike, whereas only the convention actually used retains it.

A1 Interior and boundary modes

Renda introduces a cubic cavity with mode indices l, m, n described as non-negative integers. In the continuum approximation his eq. (16) uses

$$\overline{N}(p)\,dp = \tfrac{1}{8} 4\pi p^2\,dp, \tag{A1}$$

and states that the factor $1/8$ corresponds to one octant with $l, m, n > 0$. The restriction is not merely a remark attached to the continuum estimate: it is inconsistent with the mode labelling introduced a page earlier, where l, m, n are described as non-negative integers and $\lambda_x = 2L/l$ admits $l = 0$. In the discrete calculation, $N(p)$ is then taken from OEIS A002102 at $s = p^2$, and A002102 counts all non-negative solutions of $x^2 + y^2 + z^2 = s$, including points on the coordinate planes and axes. The octant restriction of eq. (16) and the sequence used in eq. (28) therefore describe different sets.

That distinction alone is harmless asymptotically, because boundary lattice points are lower order than interior points. At small p it cannot be ignored. For the electromagnetic cavity there is an additional multiplicity issue. Renda's eq. (14) multiplies every $N(p)$ by two for the two photon polarisations. The multiplicity is indeed two for an interior mode with all three indices non-zero, but it is not two on the boundary of the octant.

Let

$$\begin{cases} N_3(s) \doteq \#\{(l,m,n) \in \mathbb{Z}^3_{>0} : l^2 + m^2 + n^2 = s\}, \\ N_2(s) \doteq \#\{(l,m) \in \mathbb{Z}^2_{>0} : l^2 + m^2 = s\}, \end{cases} \tag{A2}$$

both counting ordered tuples. For a perfectly conducting rectangular cavity the separated Maxwell solutions divide, relative to any chosen axis, into TE and TM families whose index ranges are not the same (Jackson 1999; Pozar 2011). Taking the axis to be z and writing the indices as (l, m, n), the TM family requires $l, m \geq 1$ with $n \geq 0$, while the TE family requires $n \geq 1$ with l, m not both zero. Hence a mode with all three indices non-zero is realised in both families and carries two independent polarisations; a mode with exactly one vanishing index is realised in one family only, and carries one; and a mode with two vanishing indices is realised in neither, its field being identically zero. There are three choices for the vanishing coordinate. The physical polarisation-weighted degeneracy is therefore

$$g(s) \doteq 2N_3(s) + 3N_2(s). \tag{A3}$$

Since Renda's eqs. (14) and (28) have already factored out the leading 2, the corresponding replacement inside eq. (28) is

$$N_{\mathrm{eff}}(s) \doteq N_3(s) + \frac{3}{2} N_2(s). \tag{A4}$$

By contrast, A002102 obeys

$$N_{\geq 0}(s) = N_3(s) + 3N_2(s) + 3\chi_{\square}(s), \tag{A5}$$

where

$$\chi_{\square}(s) \doteq \begin{cases} 1, & s \text{ is a perfect square}, \\ 0, & \text{otherwise}. \end{cases} \tag{A6}$$

Thus using A002102 together with the universal factor of two overweights each one-zero mode by a factor of two and also assigns weight to two-zero index triples whose cavity field vanishes.

For orientation, the first few unweighted counts are

s	1	2	3	4	5	6	7	8	9	10	11	12
$N_{\geq 0}$	3	3	1	3	6	3	0	3	6	6	3	1
N_3	0	0	1	0	0	3	0	0	3	0	3	1

The large discrepancy at small s is precisely where the claimed low-acceleration feature is generated.

A2 Continuum normalisation

The continuum normalisation itself is unchanged. Writing $s = p^2$, eq. (A1) becomes, to leading order,

$$N_{\mathrm{eff}}(s)\,ds \simeq \frac{\pi}{4}\sqrt{s}\,ds, \tag{A7}$$

because the boundary term N_2 is lower order. Substitution into Renda's eq. (28) gives

$$\frac{30\pi^3 H_0^4 c^4}{a^4} \int_0^\infty \frac{\pi}{4} \frac{s\,ds}{\mathrm{e}^{\sqrt{s}\cdot x} - 1} = 1, \qquad x \doteq \frac{\pi^2 c H_0}{a}, \tag{A8}$$

where the integral is evaluated with $\int_0^\infty y^3/(\mathrm{e}^y - 1)\,dy = \Gamma(4)\cdot\zeta(4)$. Hence the correction alters the finite-mode contribution while preserving $F(a) \to 1$ in the continuum limit.

It is worth isolating the structure that makes (A8) come out to unity exactly. Both the discrete sum and the continuum integral depend on the acceleration only through the dimensionless combination x. Writing W for whichever weight is adopted, eq. (28) of Renda (2019) may be written

$$\begin{cases} x \doteq \dfrac{\pi^2 c H_0}{a}, \\ F(x) \doteq \dfrac{30 x^4}{\pi^5} \displaystyle\sum_{s \geq 1} \frac{W(s)\sqrt{s}}{\mathrm{e}^{\sqrt{s}\cdot x} - 1}. \end{cases} \tag{A9}$$

Equation (A8) is the statement $F(x) \to 1$ as $x \to 0$. The table below is therefore one curve per weight, sampled at different x; and a constant factor in the relation between x and a slides a curve along the abscissa without altering its height. Section A4 uses this.

A2.0.1 Why a fixed Θ is sufficient here. Section 9.1 raises a genuine cosmological question about the identity and evolution of Θ, but it does not compromise the calculation of Renda (2019). Renda evaluates a single-epoch spectral function using the present value $H_0 = 2.3 \times 10^{-18}\,\mathrm{s}^{-1}$; he is not integrating the dynamics of Θ through cosmic history. With $\Theta \doteq 2c/H_0$, the scaling variable may equivalently be written

$$x \doteq \frac{\pi^2 c H_0}{a} = \frac{2\pi^2 c^2}{a \cdot \Theta}. \tag{A10}$$

Replacing the present-epoch value by the value appropriate to some other specified epoch therefore changes only the map between x and the acceleration plotted on the abscissa. It does not change the dimensionless function $F(x)$, the integer mode counts, the continuum normalisation, or the polarisation multiplicities. In particular it cannot restore the $F > 1$ overshoot once the Maxwell weighting is used. Over any observational interval relevant to a present-epoch calculation the cosmological drift of Θ is negligible. Treating Θ as fixed was therefore a perfectly sensible approximation for Renda's purpose. A different convention for its present numerical value would likewise rescale the acceleration axis, but that is logically distinct from both the factor-of-two largest-wavelength convention discussed in Section A4 and the mode-counting correction.

A3 Numerical consequence

We evaluate Renda's eq. (28) directly with $H_0 = 2.3 \times 10^{-18}\,\mathrm{s}^{-1}$, as in Renda (2019). The counting functions are generated as coefficient sequences of ordinary theta-series products and the sum is truncated at $s = 2 \times 10^6$, which is amply converged over the acceleration range shown below.

a [m s^{-2}]	A002102	N_3 only	Maxwell
5.00×10^{-10}	0.01245	0.00000034	0.0000315
8.00×10^{-10}	0.32484	0.000358	0.006869
1.20×10^{-9}	1.21468	0.010259	0.083182
2.00×10^{-9}	2.10222	0.093743	0.373270
2.42×10^{-9}	**2.16711**	0.153000	0.499146
5.00×10^{-9}	1.79414	0.439018	0.832840
6.00×10^{-9}	1.67501	0.508673	0.877806
1.00×10^{-8}	1.40879	0.674347	0.951406
2.00×10^{-8}	1.19970	0.824573	0.986943
1.00×10^{-7}	1.03832	0.962765	0.999448

The A002102 column reproduces the published ordinate $F_{\max} \simeq 2.17$ to four significant figures. With the Maxwell multiplicities of eq. (A3), the overshoot disappears: through the low-acceleration interval containing the reported critical point, $F(a)$ rises towards unity without an interior maximum. The stable acceleration inferred from the overshoot therefore does not survive the electromagnetic mode count.

The N_3-only column is included as a diagnostic because it corresponds to applying the strict-interior convention of eq. (A1) to the discrete sum as well. It also removes the peak, but it omits the legitimate one-zero electromagnetic modes and is therefore not the preferred correction.

A4 The abscissa of the published peak

A second, independent discrepancy is worth recording, because it is resolvable. Evaluating eq. (28) of Renda (2019) as printed, with A002102, reproduces the published ordinate $F \simeq 2.17$ to five significant figures but places the maximum at $x_{\max} = 2.7897$, that is $a \approx 2.439 \times 10^{-9}\,\mathrm{m\,s}^{-2}$, whereas Renda (2019) reports $a_p \approx 1.20 \times 10^{-9}\,\mathrm{m\,s}^{-2}$.

By (A9) the ordinate cannot distinguish between the two: F is a function of x alone, so a constant factor in the x–a relation moves the abscissa and leaves the height fixed. A correct ordinate with an abscissa out by two is the signature of exactly such a factor, and is not something any change of mode count can produce.

The factor can be located. With $\lambda_p = 2L/p$ and the cavity side identified with the Hubble diameter $\Theta = 2c/H_0$, the Boltzmann exponent is

$$\frac{hc}{\lambda_p \cdot kT} = \frac{h \cdot p \cdot H_0}{4} \cdot \frac{4\pi^2 c}{h \cdot a} = \frac{\pi^2 c H_0 \cdot p}{a}, \tag{A11}$$

which is eq. (28) as printed. If instead the largest admissible wavelength is taken to be twice the Hubble diameter in the sense of eq. (6) of Renda (2019), so that $\lambda_p = 4\Theta/p$, the exponent halves and the maximum moves to

$$a = \frac{\pi^2 c H_0}{2x_{\max}} = 1.220 \times 10^{-9}\,\mathrm{m\,s}^{-2}. \tag{A12}$$

That is the published abscissa to the precision at which its fig. 3 can be read. The offset is therefore consistent with a factor of two in the largest admissible wavelength between the printed equation and the evaluated one, and is independent of the mode-counting issue above.

Neither reading changes Section A3: with the Maxwell multiplicities there is no maximum at either abscissa.

Remark 2. The summation index in Renda's eqs. (25)–(28) is written $p^2 = 1$ to ∞: p^2 runs over the positive integers and $p = \sqrt{s}$ is therefore generally irrational. Restricting the sum to integer p instead gives a different calculation and fails the continuum normalisation check in eq. (A8).

A5 What is unaffected

The correction above touches one numerical claim and leaves the spine of that critique intact. The substance of his Correction 1 is that McCulloch's $F(a)$ is a linear interpolation between assumed endpoints rather than a derived quantity; that remains true. Correction 2, concerning the substitution of the peak-wavelength energy density for the integrated spectrum, is independent of the mode-counting issue and is untouched.

Neither does this appendix bear on the argument of the main text. Sections 4.1–8 concern whether an acceleration horizon performs the operation that any such spectral calculation presupposes, and are logically prior to the question of how that spectrum should be summed.